

Constraints of the MILP Model

Given parameters $f > 0$, $d, k > 0$, $b, s, g \in \mathbb{R}$, and an index set $i \in \{0, \dots, t\}$. For each time step i , the following constraints apply:

1. Big-M conditions (case distinction for $x_i + p_i$):

$$x_i + p_i \leq M(1 - \delta_i) \quad (1)$$

$$x_i + p_i \geq \varepsilon - M\delta_i \quad (2)$$

with $\delta_i \in \{0, 1\}$, a sufficiently large $M > 0$, and a small $\varepsilon > 0$.

2. Linearization of $z_i = \delta_i x_i$:

$$z_i \leq x_i - x^{\min}(1 - \delta_i) \quad (3)$$

$$z_i \geq x_i - x^{\max}(1 - \delta_i) \quad (4)$$

$$z_i \leq x^{\max} \cdot \delta_i \quad (5)$$

$$z_i \geq x^{\min} \cdot \delta_i \quad (6)$$

where $x_i \in [x^{\min}, x^{\max}]$, $z_i \in \mathbb{R}$.

3. Main constraint with factor-dependent behavior:

$$\left(1 + \frac{d}{k}\right) \cdot \sum_{i=0}^t \left(f \cdot z_i + \frac{1}{f}(x_i - z_i)\right) \leq b - s - \sum_{i=0}^t \left(f \cdot \delta_i + \frac{1}{f}(1 - \delta_i)\right) \cdot p_i \quad (7)$$

$$\left(1 + \frac{d}{k}\right) \cdot \sum_{i=0}^t \left(f \cdot z_i + \frac{1}{f}(x_i - z_i)\right) \geq -s - \sum_{i=0}^t \left(f \cdot \delta_i + \frac{1}{f}(1 - \delta_i)\right) \cdot p_i \quad (8)$$

4. Constraint for bounding the weighted quantities:

$$-g \leq \left(f \cdot \delta_i + \frac{1}{f}(1 - \delta_i)\right) \cdot (x_i + p_i) \leq g \quad \forall i = 0, \dots, t \quad (9)$$

To linearize this, we define a new auxiliary variable $w_i = \left(f \cdot \delta_i + \frac{1}{f}(1 - \delta_i)\right) \cdot (x_i + p_i)$, which is linearized using the following inequalities:

5. Linearization of w_i :

$$w_i \leq f(x_i + p_i) - M(1 - \delta_i) \quad (10)$$

$$w_i \geq f(x_i + p_i) - M(1 - \delta_i) \quad (11)$$

$$w_i \leq \frac{1}{f}(x_i + p_i) + M\delta_i \quad (12)$$

$$w_i \geq \frac{1}{f}(x_i + p_i) - M\delta_i \quad (13)$$

Together with the bound:

$$-g \leq w_i \leq g \quad \forall i = 0, \dots, t \quad (14)$$

6. Variable definitions:

$$x_i \in [x^{\min}, x^{\max}]$$

$$\delta_i \in \{0, 1\}$$

$$z_i \in \mathbb{R}$$

$$w_i \in \mathbb{R}$$